

The empirical recurrence for OEIS A304138, recovered from its tail

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Abstract

OEIS A304138 records “Empirical recurrence of order 91” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 135$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 6$, so the recurrence is proved for every $n \geq 97$. Its last coefficient is $c_{91} = -393216 \neq 0$, so no further tail satisfies anything shorter; any order-91 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A304138 is “Number of $n \times 5$ 0..1 arrays with every element unequal to 0, 2, 3 or 6 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

1, 26, 15, 20, 39, 83, 135, 356, ...

The entry states:

Empirical recurrence of order 91 (see link above)

As of the “Last modified” line on the live entry (May 07 2018 08:10:43, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 135$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 135$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 91: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 6$ is the first whose minimal order is exactly the stated 91.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 270$ exact terms from $n = 6$ on, it returns order 91 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{91} c_i a(n-i),$$

c_1	7	c_{24}	−1479593490	c_{47}	−46171495300	c_{70}	−1552862688
c_2	−19	c_{25}	2314662414	c_{48}	53379885514	c_{71}	5021451026
c_3	79	c_{26}	−3743198759	c_{49}	−68052357490	c_{72}	−5826416876
c_4	−385	c_{27}	5173290819	c_{50}	104679059229	c_{73}	2576604908
c_5	996	c_{28}	−7359292043	c_{51}	−113368426137	c_{74}	−2395030360
c_6	−2676	c_{29}	10958894358	c_{52}	110279517506	c_{75}	3034126688
c_7	9995	c_{30}	−14033786455	c_{53}	−137373647040	c_{76}	−1325722944
c_8	−24727	c_{31}	18105514118	c_{54}	143961246328	c_{77}	−81159712
c_9	54428	c_{32}	−24750936044	c_{55}	−117213199679	c_{78}	−25251776
c_{10}	−163717	c_{33}	29189658709	c_{56}	120776972968	c_{79}	97572480
c_{11}	384464	c_{34}	−33849020679	c_{57}	−122553671018	c_{80}	197202944
c_{12}	−753926	c_{35}	42221649311	c_{58}	85340556249	c_{81}	−169595904
c_{13}	1896216	c_{36}	−45640251799	c_{59}	−68210788527	c_{82}	52811264
c_{14}	−4181705	c_{37}	46711978559	c_{60}	67367734437	c_{83}	−83861760
c_{15}	7567494	c_{38}	−52319954123	c_{61}	−42156976119	c_{84}	29524480
c_{16}	−16424825	c_{39}	51802782085	c_{62}	20254633667	c_{85}	−8585216
c_{17}	33704183	c_{40}	−44847147765	c_{63}	−20453044628	c_{86}	20788224
c_{18}	−57030037	c_{41}	42189414250	c_{64}	15464853137	c_{87}	11403264
c_{19}	109458688	c_{42}	−37917756097	c_{65}	−1595042000	c_{88}	−8589312
c_{20}	−207868942	c_{43}	22659256524	c_{66}	1532115598	c_{89}	−1081344
c_{21}	329695716	c_{44}	−7726145729	c_{67}	−4304130625	c_{90}	−393216
c_{22}	−568773833	c_{45}	2012848631	c_{68}	−2570215996	c_{91}	−393216
c_{23}	997315109	c_{46}	17266822075	c_{69}	3859327520		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 97$, and it is the recurrence OEIS A304138 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 6$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{91} - c_1x^{91-1} - \dots - c_{91}$. Its constant term is $-c_{91} = 393216$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 91 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 91, so $\deg q = 91$, and it is monic. It holds from some index t on. From $\max(t, 6)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 91, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 30 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 91, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -393216 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-91 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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